

(Early vs Late Interaction in Transformer Models for Drug-Target Prediction)

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1 What question are you solving?

Drug-target interactions (DTIs) describe how a drug molecule binds to a specific protein in the body to produce a biological effect. These interactions are central to understanding how drugs work, including their intended effects and potential side effects.

In this work, we aim to investigate not only whether such interactions can be predicted, but more fundamentally: *at what stage and in what manner should drug and protein representations interact in transformer-based models?* We study whether introducing interaction early (at the embedding level) or later (after separate encoding) leads to better performance and efficiency.

2 Why is this question important to solve?

Most existing models assume that interaction should occur after separate encoding, without exploring alternative designs. However, the stage of interaction may impact both predictive performance and computational efficiency. Understanding this trade-off is important for designing more effective and scalable transformer-based DTI models.

3 What has been done already?

Darsaraee et al.(1) introduced PPB3, an online target prediction tool based on deep neural networks trained on the large and diverse ChEMBL dataset. Unlike earlier methods, PPB3 supports predictions not only for single proteins but also for protein complexes, cell lines, and whole organisms, making it one of the most comprehensive tools available. Several earlier web-based drug-target interaction prediction methods also exist, including PASS, SEA, SuperPred, TargetNet, SwissTargetPrediction, PPB, PPB2, and PLATO. Weininger (2) proposed SMILES (Simplified Molecular Input Line Entry System), a widely used representation that encodes chemical structures as strings, enabling efficient processing by computational models.

Recent work has explored advanced deep learning architectures for improving prediction performance. Lou et al(3) proposed MambaTransDTA, which combines the Mamba architecture with Transformers. This hybrid design captures both long-range dependencies and local interactions, leading to improved drug-target affinity prediction. It achieves strong performance across benchmark datasets, including Davis, KIBA, Metz, and BindingDB. Similarly, DHGT-DTI(4) integrates GraphSAGE and Graph Transformers to learn local and global structural features separately, then combines them into a unified representation for prediction.

Ibragimova et al(5) investigated the use of transfer learning from activity cliff prediction—where structurally similar compounds exhibit significantly different biological activities—to enhance drug-target interaction prediction. Finally, Abramson et al (6) presented AlphaFold 3, a diffusion-based model capable of predicting the joint 3D structure of biomolecular complexes, including proteins, nucleic acids, small molecules, ions, and modified residues, marking a significant step toward understanding molecular interactions at a structural level.

4 How are you solving this? How is it different from others' solutions?

We propose a transformer-based framework for drug-target interaction (DTI) prediction that focuses on a fundamental yet underexplored question: *at what stage and in what manner should drug and protein representations interact?*

Given a drug SMILES sequence and a protein amino acid sequence, we first convert both inputs into continuous embeddings. We then explore four architectural variants:

- **Early Concatenation:** The drug and protein embeddings are concatenated directly and passed through a shared encoder, followed by a prediction head.
- **Early Cross-Attention:** Cross-attention is applied directly on the embeddings to model interactions at a fine-grained level, and the resulting representation is passed through an encoder and prediction head.
- **Late Concatenation:** The drug and protein embeddings are first processed by separate encoders. Their encoded representations are then concatenated and passed through a joint encoder and prediction layer.
- **Late Cross-Attention:** The drug and protein embeddings are first processed independently, followed by cross-attention to fuse higher-level representations, and finally passed through a joint encoder and prediction head.

The key distinction of our work lies in not restricting interaction to a single stage. Instead, we explicitly compare early-stage and late-stage fusion strategies to understand their impact on predictive performance. Early interaction may preserve fine-grained token-level relationships, while later interaction may benefit from more abstract, high-level features.

5 What experiments will you run?

For each experiment, specify:

1. Which datasets will you use if any? Which metrics will you use to evaluate?

We will use BindingDB(7) and ChEMBL(8) as large-scale training datasets, as they contain millions of experimentally measured drug-target interaction records. Additionally, we will use Davis(9) and KIBA(10) as standardized benchmark datasets to ensure a fair comparison with existing state-of-the-art methods. For evaluation, we will use Mean Squared Error (MSE)(11) to measure prediction accuracy of binding affinities, and the Concordance Index (CI)(12) to evaluate how well the predicted rankings of interactions align with the ground truth.

2. What results do you expect to see?

We expect that models incorporating interaction between drug and protein representations will outperform those that treat them independently. In particular, we hypothesize that cross-attention based methods will achieve better performance than simple concatenation, as they can capture fine-grained relationships between drug substructures and protein residues.

Furthermore, we expect that late interaction (after separate encoding) will produce more stable and higher-level representations, while early interaction may better capture local dependencies. Among the four proposed variants, we anticipate that the model combining separate encoders with cross-attention will achieve the best overall performance.

3. Why should your solution be better than the way others have solved it?

Existing methods typically fix the interaction stage (usually after encoding), without exploring alternative design choices. Our approach systematically compares multiple interaction strategies, allowing us to identify the most effective way to model drug-protein relationships.

By explicitly studying both early and late fusion, as well as concatenation and cross-attention mechanisms, our model is able to capture both low-level token interactions and high-level semantic representations. This flexibility is expected to improve predictive performance and provide deeper insights into how drug-target interactions should be modeled in transformer-based systems.

References

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